

=== CONFIGURACION GNB (1779215560) ===

This configuration outlines how to configure the srsRAN Project gNB to create a single FDD cell
transmitting in band 3, with 10 MHz bandwidth and 15 kHz sub-carrier-spacing.
This specific example uses ZMQ in place of a USRP for the RF-frontend.
This configuration is intended to be used with srsUE.

cu_cp:

amf:

addr: 10.53.1.2 # The address or hostname of the AMF.
port: 38412
bind_addr: 10.53.1.3 # A local IP that the gNB binds to for traffic from the AMF.
supported_tracking_areas:
- tac: 7
plmn_list:
- plmn: "20893"
tai_slice_support_list:
- sst: 1 # SST value 1 corresponds to eMBB (enhanced Mobile Broadband) as per 3GPP TS 23.501 Table 5.15.
inactivity_timer: 7200 # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 7200]

ru_sdr:

device_driver: zmq # The RF driver name.
device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # ZMQ ports for gNB
srate: 11.52 # RF sample rate might need to be adjusted according to selected bandwidth.
tx_gain: 75 # Transmit gain of the RF might need to adjusted to the given situation.
rx_gain: 75 # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:

dl_arfcn: 368500 # ARFCN of the downlink carrier (center frequency).
band: 3 # The NR band.
channel_bandwidth_MHz: 10 # Bandwidth in MHz. Number of PRBs will be automatically derived.
common_scs: 15 # Subcarrier spacing in kHz used for data.
plmn: "20893" # PLMN broadcasted by the gNB.
tac: 7 # Tracking area code (needs to match the core configuration).
pdcch:
common:
ss0_index: 0 # Set search space zero index to match srsUE capabilities
coreset0_index: 6 # Set search CORESET Zero index to match srsUE capabilities for 10MHz bandwidth
dedicated:
ss2_type: common # Search Space type, has to be set to common
dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
prach:
prach_config_index: 1 # Sets PRACH config to match what is expected by srsUE
pdsch:
mcs_table: qam64 # Sets PDSCH MCS to 64 QAM
pusch:
mcs_table: qam64 # Sets PUSCH MCS to 64 QAM

log:

filename: /tmp/gnb.log # Path of the log file.
all_level: info # Logging level applied to all layers.
hex_max_size: 0

pcap:

mac_enable: false # Set to true to enable MAC-layer PCAPs.
mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
ngap_enable: false # Set to true to enable NGAP PCAPs.
ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
e2ap_enable: true # Set to true to enable E2AP PCAPs.
e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

metrics:

layers:
enable_rlc: true # Enable RLC metrics

```
enable_sched: true          # Enable DU scheduler metrics
periodicity:
  du_report_period: 1000    # Set DU statistics report period to 1s
```

=== CONFIGURACION UE (1779215560) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6 # ZMQ ports for UE (crossed)

[rat.eutra]
dl_earfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 3
nof_carriers = 1

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 00112233445566778899aabbccddeeff
imsi = 208930000000001 # IMSI with MCC 208 and MNC 93
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = internet # Common APN
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER COMPOSE (1779215560) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
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#

services:
  mongodb:
    container_name: open5gs_mongodb
    image: mongo:4.4
    networks:
      ran:
        ipv4_address: 10.53.1.100 # MongoDB IP within the Docker network
    healthcheck:
      test: echo 'db.runCommand("ping").ok' | mongo mongodb:27017/test --quiet
      interval: 3s
      timeout: 1s
      retries: 60

  5gc:
    container_name: open5gs_5gc
    build:
      context: open5gs
      target: open5gs
      args:
        OS_VERSION: "22.04"
        OPEN5GS_VERSION: "v2.7.0"
    env_file:
      - ${OPEN_5GS_ENV_FILE:-open5gs/open5gs.env}
    privileged: true
    ports:
      - "9898:9999/tcp"
    # Uncomment port to use the 5gc from outside the docker network
    # - "38412:38412/sctp" # NGAP port for gNB connection
    # - "2152:2152/udp" # GTPU port for UPF
    command: 5gc -c /open5gs/open5gs-5gc.yml # Path to the 5GC configuration file inside the container
    volumes:
      - ./open5gs-5gc.yml:/open5gs/open5gs-5gc.yml # Mount the generated 5GC config
    healthcheck:
      test: [ "CMD-SHELL", "nc -z 127.0.3.5 7777" ] # Check AMF SBI port
      interval: 3s
      timeout: 1s
      retries: 60
    networks:
      ran:
        ipv4_address: ${OPEN5GS_IP:-10.53.1.2}
    depends_on:
      - mongodb

  gnb:
```

```

container_name: srsran_gnb
# Build info
image: srsran/gnb
build:
  context: . # Assuming docker-compose.yml is in the root, and Dockerfile is in docker/
  dockerfile: docker/Dockerfile
  args:
    OS_VERSION: "24.04"
# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
  - SYS_NICE
  - CAP_SYS_PTRACE
volumes:
  # Access USB to use some SDRs
  - /dev/bus/usb/:/dev/bus/usb/
  # Sharing images between the host and the pod.
  # It's also possible to download the images inside the pod
  - /usr/share/uhd/images:/usr/share/uhd/images
  # Save logs and more into gnb-storage
  - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current network configuration creates a private network with both containers attached
# An alternative would be `network: host`. That would expose your host network into the container. It's the easiest
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.3}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
gnb_config.yml:
  file: ./gnb.yaml # Path to your desired gNB config file (generated above)
gnb_compose_config.yml:
  content: |
    cu_cp:
      amf:
        addr: ${OPEN5GS_IP:-10.53.1.2}
        bind_addr: ${GNB_IP:-10.53.1.3}
    metrics:
      autostart_stdout_metrics: true
      enable_json: true
    remote_control:
      bind_addr: 0.0.0.0
      enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:

```

```

ipam:
  driver: default
  config:
    - subnet: 172.19.1.0/24

# Open5GS 5GC Configuration (open5gs-5gc.yml) - This file should be saved separately as open5gs-5gc.yml
# and placed in the same directory as your docker-compose.yml for the volume mount to work.
#
# db_uri: mongodb://mongodb/open5gs
#
# logger:
#
# global:
#   parameter:
#     # The following 'no_' parameters are set to true by default in the template,
#     # effectively disabling 4G core functions (MME, SGWC, SGWU, PCRF, HSS)
#     # and enabling 5G core functions (NRF, SCP, AMF, SMF, UPF, AUSF, UDM, PCF, NSSF, BSF, UDR).
#     # No changes needed here to enable 5G NFs.
#     no_mme: true
#     no_sgwc: true
#     no_sgwu: true
#     no_pcrf: true
#     no_hss: true
#   #   use_mongodb_change_stream: true
#
# mme:
#   freeDiameter:
#     identity: mme.localdomain
#     realm: localdomain
#     listen_on: 127.0.3.2
#     no_fwd: true
#     load_extension:
#       - module: @build_subprojects_freeDiameter_extensions_dir@/dbg_msg_dumps.fdx
#         conf: 0x8888
#       - module: @build_subprojects_freeDiameter_extensions_dir@/dict_rfc5777.fdx
#       - module: @build_subprojects_freeDiameter_extensions_dir@/dict_mip6i.fdx
#       - module: @build_subprojects_freeDiameter_extensions_dir@/dict_nasreq.fdx
#       - module: @build_subprojects_freeDiameter_extensions_dir@/dict_nas_mipv6.fdx
#       - module: @build_subprojects_freeDiameter_extensions_dir@/dict_dcca.fdx
#       - module: @build_subprojects_freeDiameter_extensions_dir@/dict_dcca_3gpp/dict_dcca_3gpp.fdx
#     connect:
#       - identity: hss.localdomain
#         address: 127.0.3.8
#
# slap:
#   server:
#     - address: 127.0.3.2
#
# gtpc:
#   server:
#     - address: 127.0.3.2
#   client:
#     sgwc:
#       - address: 127.0.3.3
#     smf:
#       - address: 127.0.3.4
#
# metrics:
#   server:
#     - address: 127.0.3.2
#     port: 9090
#
# gummei:
#   plmn_id:
#     mcc: 208
#     mnc: 093
#   mme_gid: 2
#   mme_code: 1
#
# tai:
#   plmn_id:
#     mcc: 208

```

```
# mnc: 093
# tac: 1
# security:
#   integrity_order : [ EIA2, EIA1, EIA0 ]
#   ciphering_order : [ EEA0, EEA1, EEA2 ]
# network_name:
#   full: Open5GS
# time:
#   t3412:
#     value: 3240
#
# sgwc:
#   gtpc:
#     server:
#       - address: 127.0.3.3
#   pfcp:
#     server:
#       - address: 127.0.3.3
#     client:
#       sgwu:
#         - address: 127.0.3.6
#
# smf:
#   sbi:
#     server:
#       - address: 127.0.3.4
#       - port: 7777
#       - address: smf.5gc.mnc093.mcc208.3gppnetwork.org
#     client:
#       scp:
#         - uri: http://127.0.3.200:7777
#   pfcp:
#     server:
#       - address: 127.0.3.4
#     client:
#       upf:
#         - address: 127.0.3.7
#   gtpc:
#     server:
#       - address: 127.0.3.4
#   gtpu:
#     server:
#       - address: 127.0.3.4
#   metrics:
#     server:
#       - address: 127.0.3.4
#       port: 9090
#   session:
#     - subnet: 10.47.0.0/16
#     gateway: 10.47.0.1
#     - subnet: 2001:db8:face::/48
#     gateway: 2001:db8:face::1
#   dns:
#     - 8.8.8.8
#     - 8.8.4.4
#     - 2001:4860:4860::8888
#     - 2001:4860:4860::8844
#   mtu: 1400
#   freeDiameter:
#     identity: smf.localdomain
#     realm: localdomain
#     listen_on: 127.0.3.4
#     no_fwd: true
#     load_extension:
#       - module: @build_subprojects_freeDiameter_extensions_dir@/dbg_msg_dumps.fdx
#       conf: 0x8888
#       - module: @build_subprojects_freeDiameter_extensions_dir@/dict_rfc5777.fdx
#       - module: @build_subprojects_freeDiameter_extensions_dir@/dict_mip6i.fdx
```

```

# - module: @build_subprojects_freeDiameter_extensions_dir@/dict_nasreq.fdx
# - module: @build_subprojects_freeDiameter_extensions_dir@/dict_nas_mipv6.fdx
# - module: @build_subprojects_freeDiameter_extensions_dir@/dict_dcca.fdx
# - module: @build_subprojects_freeDiameter_extensions_dir@/dict_dcca_3gpp/dict_dcca_3gpp.fdx
# connect:
# - identity: pcrf.localdomain
#   address: 127.0.3.9
#
# amf:
#   sbi:
#     server:
#       - address: 127.0.3.5
#         port: 7777
#     client:
#       scp:
#         - uri: http://127.0.3.200:7777
#   ngap:
#     server:
#       - address: 10.53.1.2 # AMF IP for gNB connection
#   metrics:
#     server:
#       - address: 127.0.3.5
#         port: 9090
#   access_control:
#     - plmn_id:
#         mcc: 208
#         mnc: 093
#   guami:
#     - plmn_id:
#         mcc: 208
#         mnc: 093
#       amf_id:
#         region: 2
#         set: 1
#   tai:
#     - plmn_id:
#         mcc: 208
#         mnc: 093
#       tac: 1
#   plmn_support:
#     - plmn_id:
#         mcc: 208
#         mnc: 093
#   s_nssai:
#     - sst: 1 # SST value 1 corresponds to eMBB (enhanced Mobile Broadband)
#   security:
#     integrity_order : [ NIA2, NIA1, NIA0 ]
#     ciphering_order : [ NEA0, NEA1, NEA2 ]
#   network_name:
#     full: Open5GS
#   amf_name: open5gs-amf0
#   time:
#     t3512:
#       value: 540      # 9 minutes * 60 = 540 seconds
#
# sgwu:
#   pfcp:
#     server:
#       - address: 127.0.3.6
#   gtpu:
#     server:
#       - address: 127.0.3.6
#
# upf:
#   pfcp:
#     server:
#       - address: 127.0.3.7
#   gtpu:

```

```

#   server:
#       - address: 127.0.3.7
# session:
#       - subnet: 10.47.0.0/16
#         gateway: 10.47.0.1
#         dev: ogstun # TUN device for UPF
#       - subnet: 2001:db8:face::/48
#         gateway: 2001:db8:face::1
#         dev: ogstun
# metrics:
#   server:
#       - address: 127.0.3.7
#         port: 9090
#
# hss:
#   freeDiameter:
#       identity: hss.localdomain
#       realm: localdomain
#       listen_on: 127.0.3.8
#       no_fwd: true
#       load_extension:
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dbg_msg_dumps.fdx
#             conf: 0x8888
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_rfc5777.fdx
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_mip6i.fdx
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_nasreq.fdx
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_nas_mipv6.fdx
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_dcca.fdx
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_dcca_3gpp/dict_dcca_3gpp.fdx
#       connect:
#           - identity: mme.localdomain
#             address: 127.0.3.2
# pcrf:
#   freeDiameter:
#       identity: pcrf.localdomain
#       realm: localdomain
#       listen_on: 127.0.3.9
#       no_fwd: true
#       load_extension:
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dbg_msg_dumps.fdx
#             conf: 0x8888
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_rfc5777.fdx
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_mip6i.fdx
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_nasreq.fdx
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_nas_mipv6.fdx
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_dcca.fdx
#           - module: @build_subprojects_freeDiameter_extensions_dir@/dict_dcca_3gpp/dict_dcca_3gpp.fdx
#       connect:
#           - identity: smf.localdomain
#             address: 127.0.3.4
#
# nrf:
#   serving:
#       - plmn_id:
#           mcc: 208
#           mnc: 093
#   sbi:
#       server:
#           - address: 127.0.3.10
#             port: 7777
#           - address: nrf.5gc.mnc093.mcc208.3gppnetwork.org
#
# scp:
#   sbi:
#       server:
#           - address: 127.0.3.200
#             port: 7777
#       client:

```

```

# nrf:
#   - uri: http://nrf.5gc.mnc093.mcc208.3gppnetwork.org
#
# # SEPP section removed as per request
#
# ausf:
#   sbi:
#     server:
#       - address: 127.0.3.11
#       port: 7777
#       - address: ausf.5gc.mnc093.mcc208.3gppnetwork.org
#     client:
#     scp:
#       - uri: http://127.0.3.200:7777
#
# udm:
#   hnet:
#     - id: 1
#       scheme: 1
#       key: @build_configs_dir@/open5gs/hnet/curve25519-1.key
#     - id: 2
#       scheme: 2
#       key: @build_configs_dir@/open5gs/hnet/secp256r1-2.key
#   sbi:
#     server:
#       - address: 127.0.3.12
#       port: 7777
#       - address: udm.5gc.mnc093.mcc208.3gppnetwork.org
#     client:
#     scp:
#       - uri: http://127.0.3.200:7777
#
# pcf:
#   sbi:
#     server:
#       - address: 127.0.3.13
#       port: 7777
#     client:
#     scp:
#       - uri: http://127.0.3.200:7777
#   metrics:
#     server:
#       - address: 127.0.3.13
#       port: 9090
#   policy:
#     - supi_range:
#       - 208930000000001-208930999999999 # SUPI range for MCC 208, MNC 093
#     slice:
#       - sst: 1 # SST value 1 (eMBB)
#         default_indicator: true
#         session:
#           - name: internet
#             type: 3 # 1:IPv4, 2:IPv6, 3:IPv4v6
#             ambr:
#               downlink:
#                 value: 1
#                 unit: 3 # 0:bps, 1:Kbps, 2:Mbps, 3:Gbps, 4:Tbps
#               uplink:
#                 value: 1
#                 unit: 3
#             qos:
#               index: 9
#             arp:
#               priority_level: 8
#               pre_emption_vulnerability: 1
#               pre_emption_capability: 1
#           - name: ims
#             type: 3

```

```

#         ambr:
#             downlink:
#                 value: 1
#                 unit: 3
#             uplink:
#                 value: 1
#                 unit: 3
#         qos:
#             index: 5
#             arp:
#                 priority_level: 1
#                 pre_emption_vulnerability: 1
#                 pre_emption_capability: 1
#         pcc_rule:
#             - qos:
#                 index: 1
#                 arp:
#                     priority_level: 1
#                     pre_emption_vulnerability: 1
#                     pre_emption_capability: 1
#                 mbr:
#                     downlink:
#                         value: 82
#                         unit: 1
#                     uplink:
#                         value: 82
#                         unit: 1
#                 gbr:
#                     downlink:
#                         value: 82
#                         unit: 1
#                     uplink:
#                         value: 82
#                         unit: 1
#             - qos:
#                 index: 2
#                 arp:
#                     priority_level: 4
#                     pre_emption_vulnerability: 2
#                     pre_emption_capability: 2
#                 mbr:
#                     downlink:
#                         value: 802
#                         unit: 1
#                     uplink:
#                         value: 802
#                         unit: 1
#                 gbr:
#                     downlink:
#                         value: 802
#                         unit: 1
#                     uplink:
#                         value: 802
#                         unit: 1
# nssf:
#     sbi:
#         server:
#             - address: 127.0.3.14
#             - port: 7777
#             - address: nssf.5gc.mnc093.mcc208.3gppnetwork.org
#     client:
#         scp:
#             - uri: http://127.0.3.200:7777
#         nsi:
#             - uri: http://127.0.3.10:7777
#         s_nssai:
#             sst: 1
# bsf:

```

```
#  sbi:
#    server:
#      - address: 127.0.3.15
#        port: 7777
#    client:
#      scp:
#        - uri: http://127.0.3.200:7777
#
# udr:
#  sbi:
#    server:
#      - address: 127.0.3.20
#        port: 7777
#    client:
#      scp:
#        - uri: http://127.0.3.200:7777
```